



American International University- Bangladesh (AIUB)
Faculty of Engineering (FE)
Department of Electrical and Electronic Engineering (EEE)

Course Project Report Outline (Digital Logic and Circuits Lab)

Group 02_DLC LAB_X_Project Report

1. Download the template for project report writing from portal.
2. Title, Abstract (at least 150 words but not more than 300 words) and Keywords (3-6 keywords separated by a comma) [3 marks]

3. Introduction

3.1. Background of Study and Motivation	[1 mark]
3.2. Project Objectives	[1 mark]
3.3. A brief Outline of the Report	[Optional]

4. Literature Review (*At least 5 project-related published journal papers or conference proceedings within the year 2019 to 2024*) [3 marks]

5. Methodology and Modeling

5.1. Introduction	[1 mark]
5.2. Working Principle of the Proposed Project	[1 mark]
5.3. Description of the Components	[0.5 mark]
5.4. Test/Experimental Setup	[0.5 mark]

6. Results and Discussions

6.1. Simulation/Numerical Analysis	[2 mark]
6.2. Measured response/Experimental Results	[2 mark]
6.3. Comparison between Numerical and Experimental Results	[1 mark]
6.4. Cost Analysis	[1 mark]
6.5. Limitations in the Project	[1 mark]

7. Conclusion and Future Endeavors [1 mark]

References [1 mark]

Appendix (if any, optional)

Assessment Materials and Marks Allocation:

COs	Assessment Materials	POIs	Marks
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CO2	Course Project Report (<i>Analyze a combinational/sequential logic circuit through appropriate survey of research literature to provide valid conclusion acknowledging the limitations.</i>)	P.d.2.C4	20
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COs	Excellent to Proficient [16-20]	Good [11-15]	Acceptable [5-10]	Unacceptable [1-4]	No Response [0]	Secured Marks
CO2 P.d.2.C4	The outcome of the project demonstrates a course project using logic ICs, transistors, switches, display devices, etc. that can solve a complex engineering problem in the electrical and electronic engineering discipline through appropriate research.	The outcome of the project somewhat demonstrates a course project using logic ICs, transistors, switches, display devices, etc., and also somewhat solves a complex engineering problem in the electrical and electronic engineering discipline through some research.	The outcome of the project demonstrates a course project using logic ICs, transistors, switches, display devices, etc. but cannot solve a complex engineering problem properly in the electrical and electronic engineering discipline through appropriate research.	The outcome of the project does not demonstrate a course project using logic ICs, transistors, switches, display devices, etc. also could not solve a complex engineering problem in the electrical and electronic engineering discipline through appropriate research.	No Response at all	
Comments					Total Marks (20)	

SL #	ID	Student Name	Contribution	Marks
1	23-55556-3	MD. SHAFAYAT HOSSAIN SHAMIM		
2	23-54150-3	APURBA SAHA		
3	23-55602-3	GAZI SAJID SALMAN		
4	2456903-1	TAMANNA BINTE WALIUR		

Automatic Street Light System Using LDR and NE555

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Abstract— Automatic street lighting systems are essential for energy conservation and smart city development. This project presents a simple and cost-effective automatic street light control system using a Light Dependent Resistor (LDR), NE555 timer IC, and transistor switching. The system automatically turns ON the street light during darkness and turns it OFF during daylight without human intervention. The LDR senses ambient light intensity and converts it into an electrical signal. This signal is processed using a voltage divider network and fed into the 555 timer IC configured as a comparator. Based on the threshold voltage, the output drives a BC547 transistor, which controls the LED acting as the street light. The use of a potentiometer allows adjustment of sensitivity according to environmental conditions. The proposed system reduces energy consumption, minimizes manual effort, and increases efficiency. This project is suitable for real-world implementation in highways, gardens, and smart lighting systems.

Index Terms— Automatic lighting, LDR sensor, 555 timer IC, energy saving, light control system.

I. INTRODUCTION

A. Background of Study and Motivation

Street lighting is a crucial part of urban infrastructure [1]. Conventional street lights remain ON for long durations, leading to unnecessary power consumption [1]. With increasing demand for energy efficiency, automatic systems are required [2]. This project aims to design an intelligent lighting system that operates based on environmental light conditions [2].

B. Project Objective

1. To design an automatic street light system using simple components.
2. To reduce energy consumption.
3. To understand the working of LDR and 555 timer IC.
4. To implement a cost-effective and efficient lighting system.

C. Outline of the Report

This report includes introduction, literature review, methodology, results, and conclusion of the automatic street light system.

II. LITERATURE REVIEW

Automatic street lighting systems have been widely studied in recent years to improve energy efficiency and reduce manual operation. A study in 2019 presented the design and calibration of an automatic street light controller using a Light Dependent Resistor (LDR), showing that such systems can reduce energy consumption by more than 50% [3]. This demonstrates the effectiveness of simple sensor-based lighting control.

In 2020, a conference paper proposed an energy-efficient street lighting system using ultrasonic sensors, which dynamically controls lighting based on vehicle presence, thereby minimizing unnecessary power usage [4]. Similarly, another 2020 research work introduced IoT-based smart street lighting systems capable of monitoring and controlling street lights remotely, enhancing efficiency and fault detection capabilities [5].

Further improvements have been made using intelligent and smart city approaches. A 2020 study on intelligent street lighting systems highlighted the use of environmental sensing to automatically adjust lighting conditions, improving both safety and energy efficiency [6]. Another research paper in 2021 proposed a smart street light management system using IoT and LDR sensors, enabling automatic brightness control and reduced electricity consumption [7].

More recently, a 2023 study focused on LDR-based automatic street lighting systems, emphasizing cost-effective implementation and reliable operation without the need for complex microcontrollers [8]. Additionally, a 2024 study integrated LDR and PIR sensors with Arduino to create an advanced automated lighting system that responds to both ambient light and motion, significantly improving energy savings and operational intelligence [9].

From the above studies, it is clear that automatic street lighting systems have evolved from simple LDR-based designs to advanced IoT-enabled smart systems. However, simple analog systems using components like LDR and 555 timer IC remain important due to their low cost, simplicity, and ease of implementation, especially for basic applications.

III. METHODOLOGY AND MODELING

A. Introduction

The methodology and modeling of this project were focused on developing an automatic street light system that operates based on ambient light intensity. A structured approach was followed to design, analyze, and implement the system using basic electronic components. The modeling part involved understanding how the sensing, control, and switching sections interact with each other to achieve automatic operation.

At the beginning, the behavior of the light sensing element was studied to determine how changes in light intensity affect the circuit. Based on this, a control mechanism was designed to process the input signal and generate a suitable output. The overall system was modeled as a combination of three main stages: sensing, signal processing, and output switching.

After the modeling phase, the circuit was physically implemented on a breadboard by properly connecting all components. The system was then adjusted to achieve the desired sensitivity and tested under different lighting conditions. This combined methodology and modeling approach ensured that the project was both theoretically sound and practically functional.

B. Working Principle of the Proposed Project

The system operates on the principle of light sensing and threshold comparison.

1. The LDR changes its resistance depending on light intensity:
 - High resistance in darkness
 - Low resistance in bright light
2. The LDR is connected in a voltage divider configuration with a 10k Ω potentiometer, which allows adjustment of the light sensitivity threshold.
3. The voltage from this divider is fed into the 555 timer IC, configured as a comparator (or switching circuit).
4. When light intensity falls below a set level (evening/night):
 - The LDR resistance increases
 - The input voltage changes
 - The 555 timer output goes HIGH
 - This activates the BC547 transistor
5. The transistor acts as a switch:
 - When turned ON, current flows through the LED.
 - The LED lights up (simulating a street light).
6. During daylight:
 - LDR resistance decreases
 - Output of 555 timer becomes LOW
 - Transistor turns OFF
 - LED remains OFF

C. Description of Components

1. LDR (Light Dependent Resistor)
 - Senses ambient light
 - Resistance varies with light intensity
2. 555 Timer IC
 - Works as a comparator/switching device
 - Processes input voltage from LDR circuit
 - Produces HIGH or LOW output based on threshold
3. BC547 Transistor
 - Acts as an electronic switch
 - Amplifies current from 555 timer output
 - Controls LED operation
4. Potentiometer (10k Ω)
 - Adjusts sensitivity of the system
 - Sets the threshold light level for switching
5. Resistors (100k Ω , 1k Ω , 200 Ω)
 - 100k Ω : Used in voltage divider with LDR
 - 1k Ω : Limits base current of transistor
 - 200 Ω : Protects LED from excessive current
6. Capacitors (10 μ F, 0.01 μ F)
 - 10 μ F: Stabilizes voltage and timing response
 - 0.01 μ F: Filters noise and ensures stable operation of the 555 timer
7. LED
 - Acts as the output light (street light model)
8. Breadboard and Connecting Wires
 - Used to assemble and connect all components without soldering

D. Experimental Setup

1. All the required components were first collected and arranged properly on a breadboard. The connections were made carefully to ensure proper circuit operation and to avoid any short circuits.
2. The LDR and the 10k Ω potentiometer were connected together to form a voltage divider circuit, and this arrangement was used to sense the ambient light intensity.
3. The capacitors (10 μ F and 0.01 μ F) were connected at appropriate positions in the circuit to stabilize the voltage and to reduce noise, ensuring smooth operation of the system. The output pin of the 555 timer IC was connected to the base of the BC547 transistor through a 1k Ω resistor, which was used to limit the base current.
4. The LED was connected in series with a 200 Ω resistor and was attached to the collector terminal of the transistor, while the emitter of the transistor was connected to ground. This configuration allowed the transistor to act as a switch for controlling the LED.
5. A 12V power supply was provided to the entire circuit to operate all components. After completing the connections, the potentiometer was adjusted to set the desired sensitivity level of the system.
6. Finally, the circuit was tested by covering and exposing the LDR to different light conditions, and the response of the LED was observed to verify proper functionality.

IV. RESULTS AND DISCUSSION

A. Simulation

The given circuit simulation represents an automatic light-controlled switching system using an LDR, 555 timer IC, and a BC547 transistor. In this simulation, the LDR (Light Dependent Resistor) and 10k Ω potentiometer are connected as a voltage divider. This divider produces a variable voltage depending on the surrounding light intensity. When light intensity changes, the resistance of the LDR also changes, which directly affects the input voltage of the 555 timer IC. The 555 timer (U2) is configured to respond to this varying voltage. When the light level crosses a certain threshold set by the potentiometer, the output pin of the 555 timer changes its state (HIGH or LOW). This output is connected to the base of the BC547 transistor (Q1) through a 1k Ω resistor (R1), which limits base current and protects the transistor. The transistor acts as an electronic switch. At the output stage, the LED with a 220 Ω resistor (R4) is connected to the collector of the transistor. When the transistor turns ON, current flows through the LED and it lights up; when it is OFF, the LED turns off. The capacitors (10 μ F and 0.01 μ F) are used for smoothing voltage fluctuations and stabilizing the triggering operation of the 555 timer.

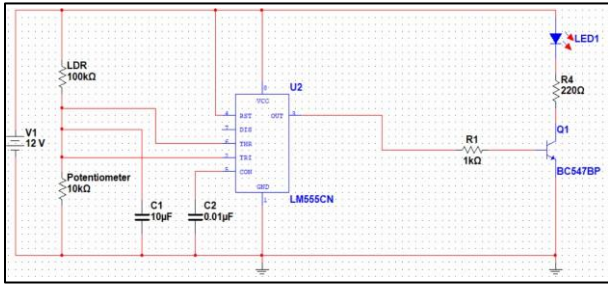


Figure-1: Simulation of Automatic Street Light Project System

B. Experimental Result

After building the circuit on a breadboard and applying a 12V DC supply, the system was tested under different light conditions:

- In bright light, the resistance of the LDR decreased, causing the 555 timer output to remain in a state that kept the transistor OFF. As a result, the LED remained OFF.
- In low light or darkness, the resistance of the LDR increased, changing the voltage at the trigger input. This caused the 555 timer output to activate the transistor, and the LED turned ON.
- By adjusting the 10k Ω potentiometer, the sensitivity of the switching point was successfully controlled.

The circuit responded smoothly and the LED switching was clearly visible during testing.

C. Comparison Between Numerical and Experimental Results

Simulation Result	Experimental Result
1. Instant ON/OFF switching based on light level	1. Slight delay due to real component behavior.
2. Fully stable and ideal switching.	2. Minor fluctuations due to noise and wiring.
3. Threshold control exact and precise via potentiometer.	3. Adjustable but slightly sensitive in real setup.
4. Ideal behavior assumed.	4. Real-world variations in LDR and transistor response.
5. Perfect logical operation.	5. Very close to simulation with small practical deviations.

D. Cost Analysis

Component	Approx Cost (Tk)
LDR	75
555 Timer IC	20
BC547 and Potentiometer	50
Resistors and Capacitors	40
LED and 9v Battery	100
Breadboard and Wire	100
Total	400 -/-

E. Limitations in The Project

1. The system is highly sensitive to environmental conditions such as dust, shadows, and unwanted light sources, which may affect the accuracy of the LDR sensor.
2. It is not suitable for high-power street lighting applications without using additional components like relays or power drivers.
3. The system provides limited accuracy and reliability compared to digital or microcontroller-based smart lighting systems.
4. It does not include any intelligent control or programmability, so it cannot adapt dynamically to different environmental conditions.
5. The circuit is affected by noise and voltage fluctuations, which may lead to unstable operation in some cases.
6. The system lacks advanced features such as motion detection, remote monitoring, and IoT integration, reducing overall efficiency.

CONCLUSION AND FUTURE ENDEAVORS

The Automatic Street Light System using LDR and 555 Timer IC successfully demonstrates a simple and cost-effective approach for energy-efficient lighting control. The system automatically switches the light ON in darkness and OFF in daylight with reliable performance in both simulation and practical implementation. It meets the basic objectives of reducing manual effort and saving energy using easily available analog components. In future, the system can be improved by integrating microcontrollers or IoT modules for smart control and remote monitoring. It can also be enhanced using solar power systems, motion sensors, and high-power LED drivers to make it more suitable for real-world street lighting applications.

ACKNOWLEDGMENT

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Appendix

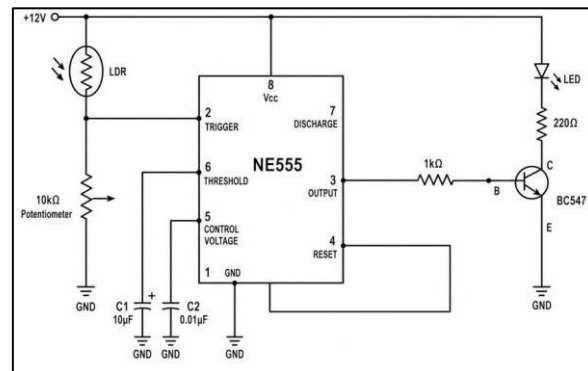


Figure-1: Circuit Diagram

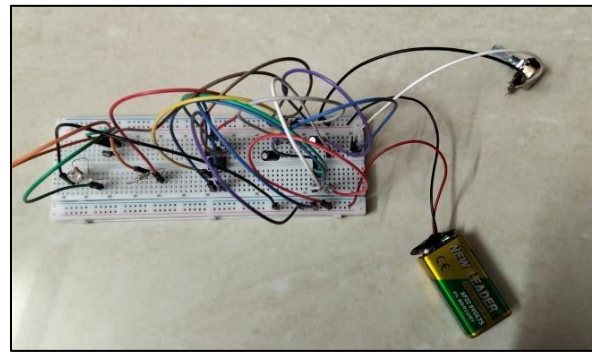


Figure-2: Our Project